

1.3 GHz Amplifier Design

EE 4140

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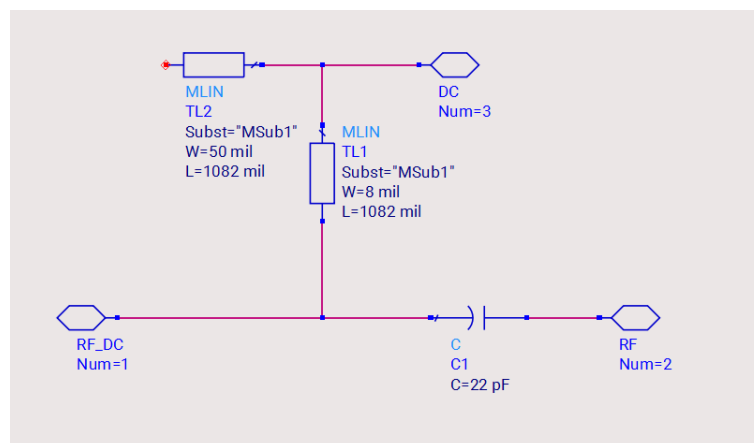
Introduction

In this lab, we were tasked with designing a microwave amplifier, optimizing for maximum gain, while meeting specifications presented in the lab manual. Our chosen design frequency for this amplifier was 1.3 GHz. At this frequency, according to the lab manual, we were required to design for a gain of at least $(21 - 8 \times 1.3) = 10.6$ dB, with $|S_{11}|$ and $|S_{22}|$ values of less than -15 dB. Along with those requirements, the amplifier must be unconditionally stable from 10 MHz through 6 GHz. The transistor is biased with a collector current of 10mA and a collector-emitter voltage of 8V.

In order to meet these specifications, our amplifier had to not only have a well-designed DC bias interface, but also needed effective stability and matching networks to compensate for those requirements in S-parameters and stability. The fractional bandwidth also must be $>5\%$, which doesn't seem like too difficult of a requirement to meet, so we decided to not really worry about that requirement.

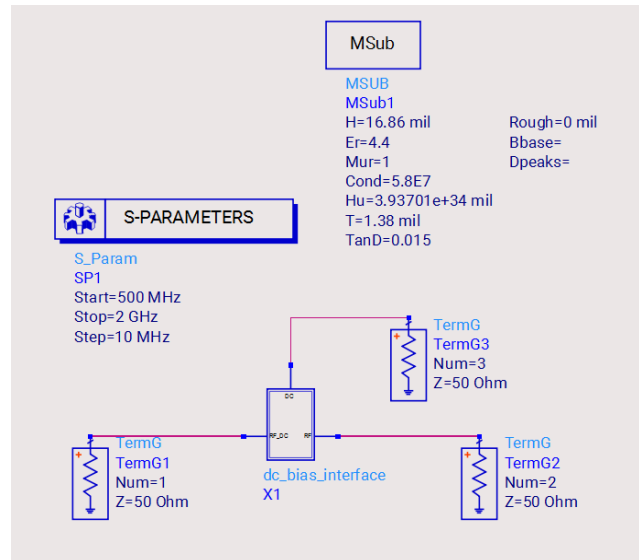
DC Bias Tee

First, we had to create a DC bias interface, which allows a DC bias to be added to the transistor without interfering with the RF signal path. In order to do this, we add a DC block (capacitor) in series with the RF signal input, and we add an RF open-circuit between the DC source and the RF signal path. To realize the RF-open, we could have chosen between an RF choke or a quarter-wave transformer, so we chose a quarter-wave transformer because they look cooler. To better realize the RF-open, we attached an open-circuited quarter-wave transmission line on the DC side, which forces the DC to be an RF-short, forcing the end connected to the RF signal path to be an RF-open. 8 mils was selected to be the width of the short-circuited transmission line, as the higher characteristic impedance (from the narrower trace) would present a better open-circuit. 1082.07 mils is quarter-wave length, and the mostly idealized version (still using MLINs however) of this network is shown below:

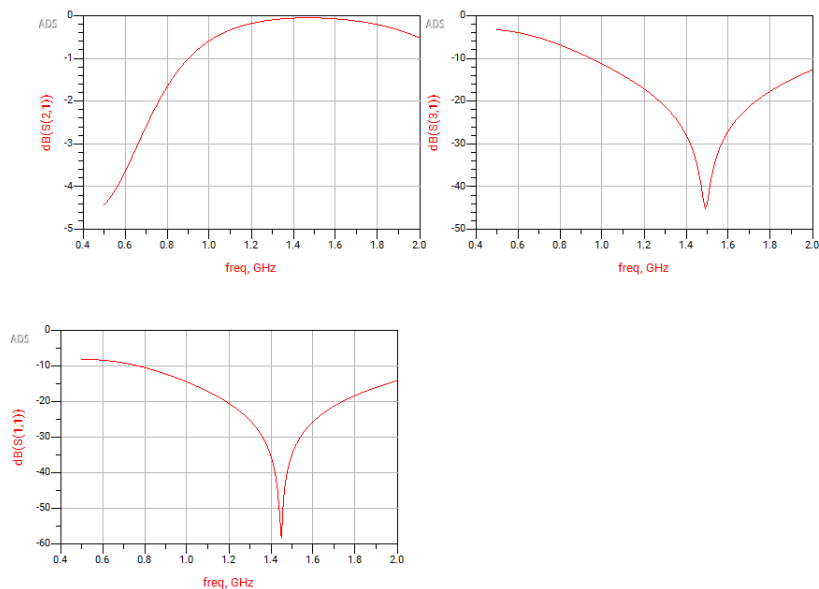


First-pass Attempt at Bias Tee

After capturing the ideal schematic, we tested it out using the testbench below, to check for S-parameters. The results of this test are shown below the testbench.



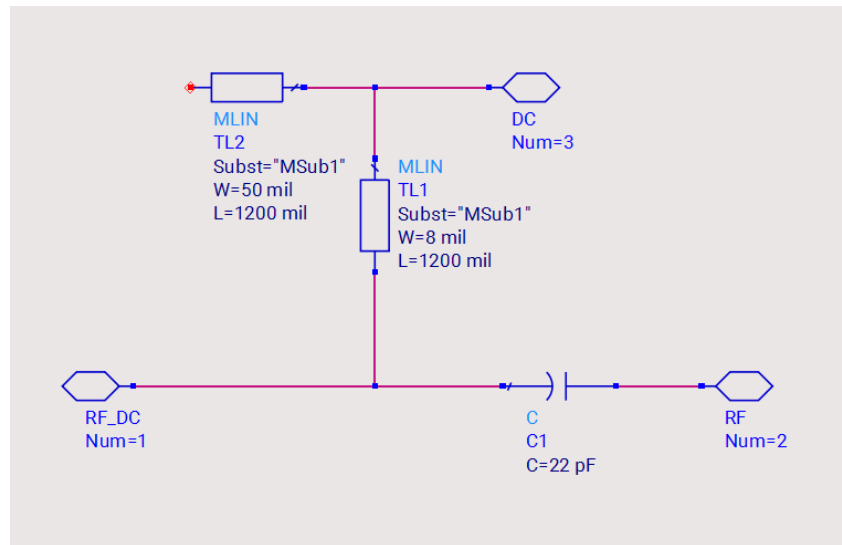
Bias Tee Testbench



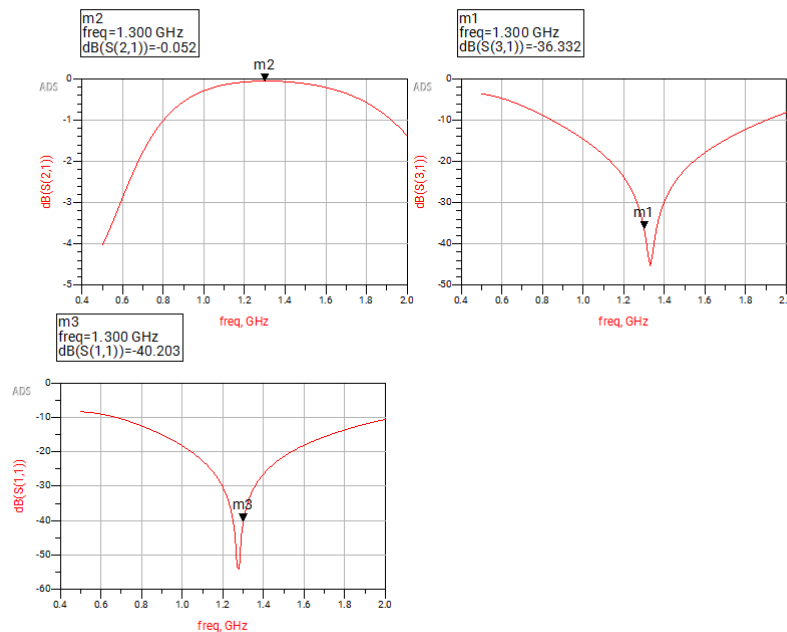
Bias Tee Testbench Results

From these results, it can be seen that the resonant frequency of this network is a little bit higher than our targeted 1.3 GHz. To remediate this, we made the quarter-wave transformers a little bit longer, as that should account for the surrounding non-idealities, and the increase in length should correspond to a smaller frequency, i.e. the one which we are designing for. The increased length can be seen in the screenshot below, and the subsequent performance improvement can be

seen below that, with the peaks and valleys of S_{21} and S_{11}/S_{31} respectively being closer to 1.3 GHz, with each of those parameters being well within their required ranges.



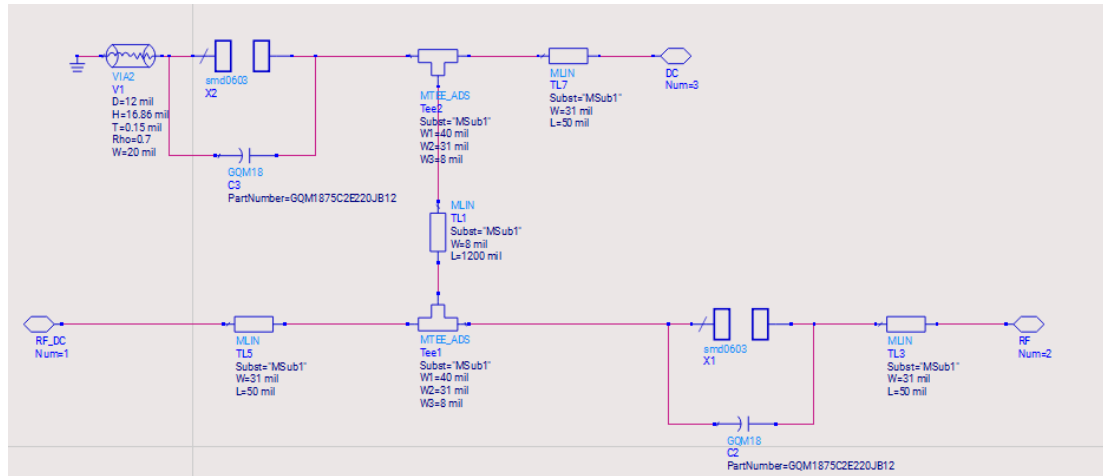
Improved Bias Tee



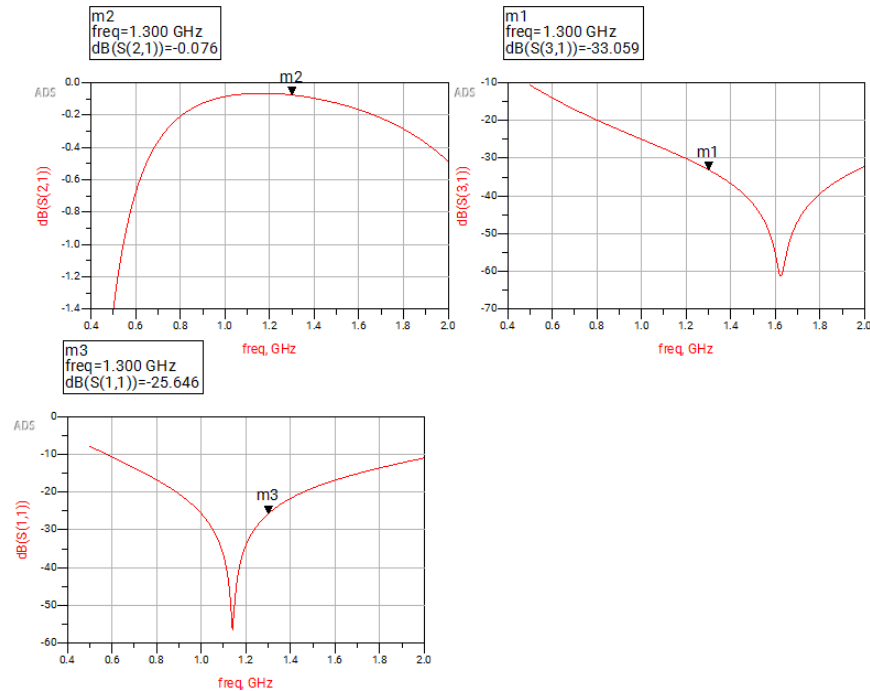
Improved Bias Tee Results

After we had the more idealized version of the bias tee, we continued to add non-idealities. During this process, we realized that having a fully distributed bias tee would be incredibly large at such a relatively low frequency, as the quarter-wave transformers were over an inch long. To reduce the size, we replaced the DC-side quarter-wave transformer with a somewhat large capacitor (22pF). This would replace the effective short caused by the transformer with a

different type of RF short. The results of these changes can be seen below in the finished schematic and the S-parameter simulation, showing acceptable (rather good) results.

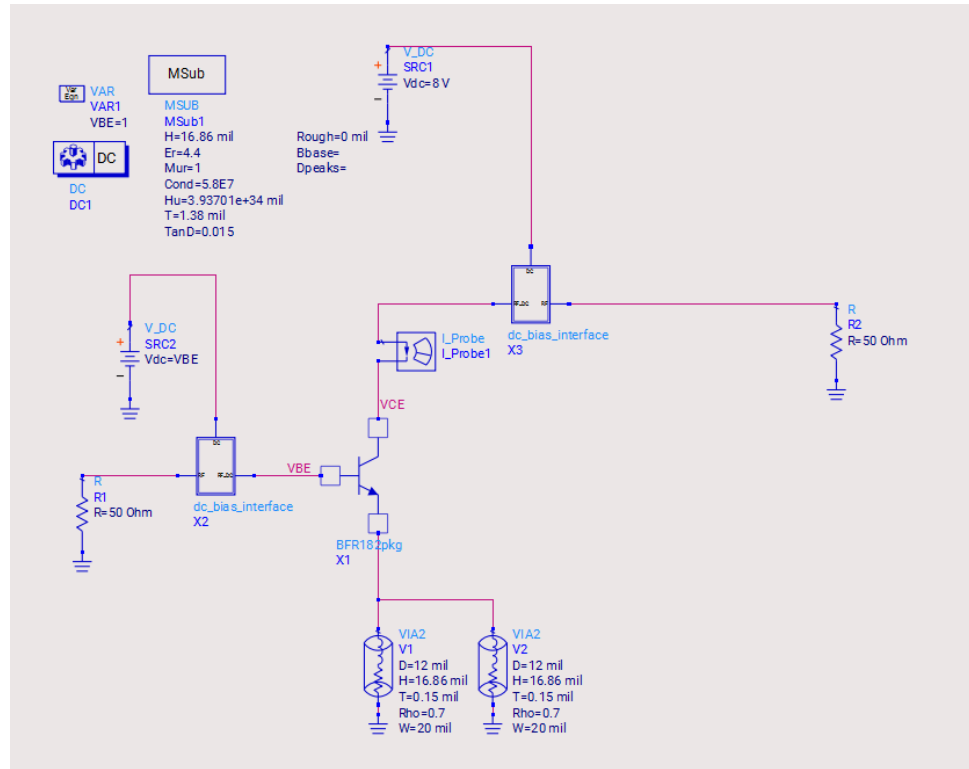


Improved Model of the Bias Tee (added non-ideal components)

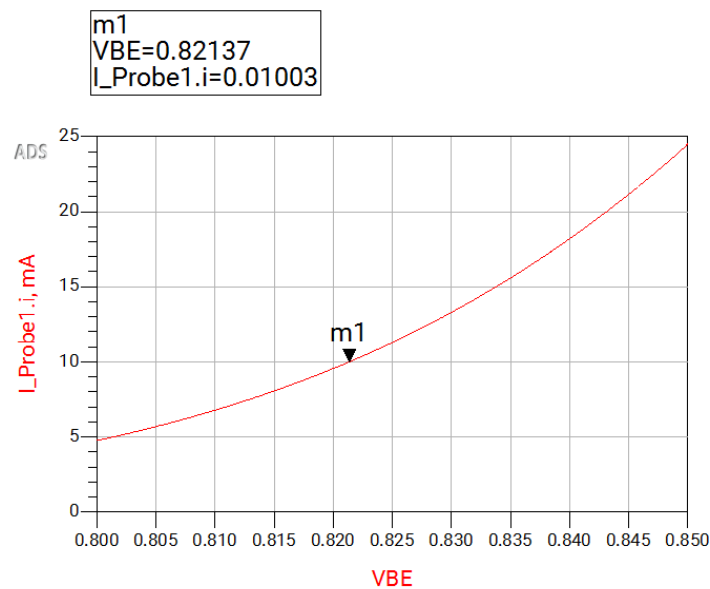


Results of More Accurate Bias Tee Simulation

Now that the bias tee was designed, we had to implement a few instances of it into the schematic with the BJT to set the DC bias point. The V_{CE} needed to be set to 8V, and we had to sweep the V_{BE} point to ensure 10mA of current was following into the collector. This testbench can be seen below, with the correct bias point found of 0.82137V for V_{BE} .



Bias Tees in Use

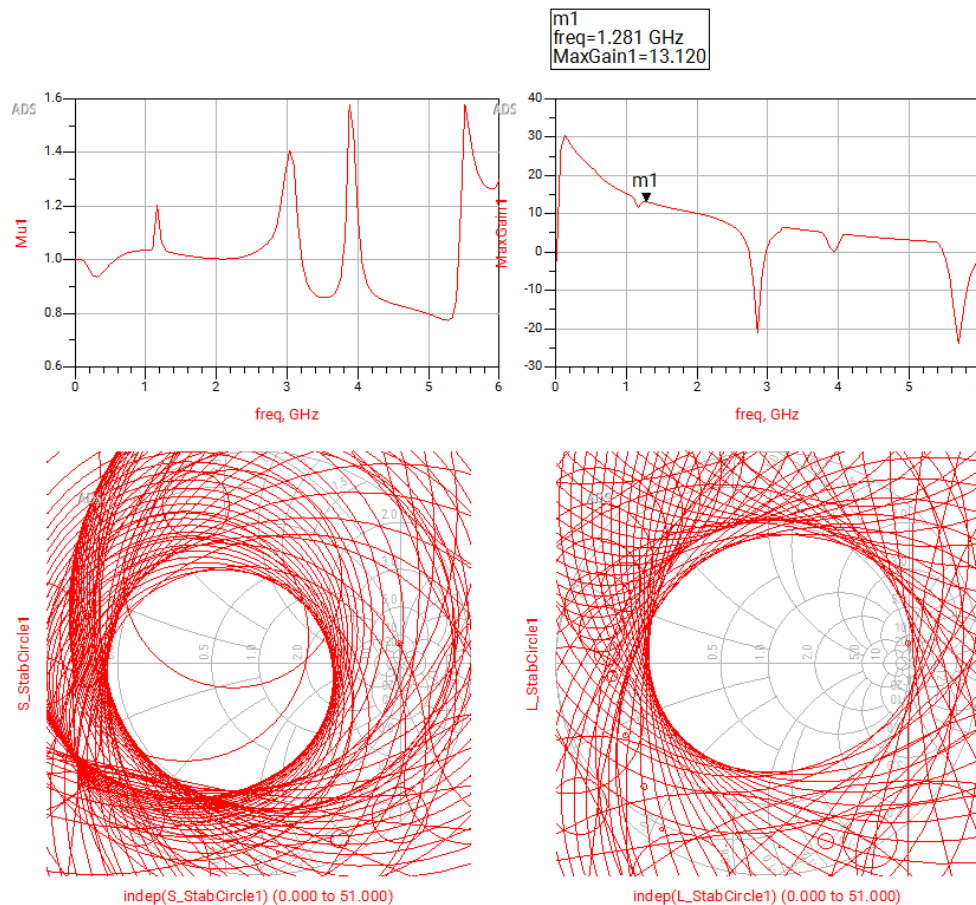


DC Bias Current Curve

Stabilization

Now that the BJT was properly biased, it was time to ensure that it exhibited unconditional stability across all frequencies (10MHz to 6GHz). The first step towards achieving that was checking the current μ factor over that range. From there, the stability circles at different frequencies can be used to facilitate our stabilization process. These pieces of information, along

with maximum gain over frequency, can be seen below, showing that our amplifier was initially unconditionally stable at our design frequency.

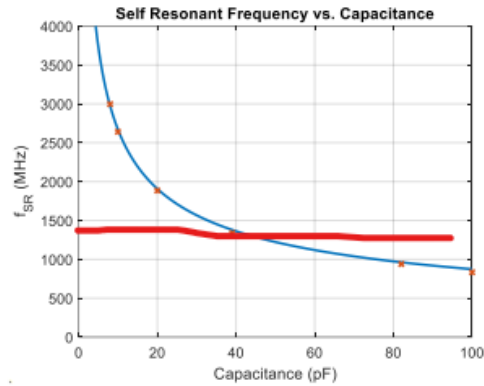


Initial Stability of Amplifier over Frequency

In order to stabilize this circuit, we utilized information from the stability circles to begin the process. The first thing we noticed was that the open-circuit on the source stability side was in the unstable region (the stability regions are not pictured, but we listed them separately). This was an easy first step towards stability: adding a shunt resistor at the source side.

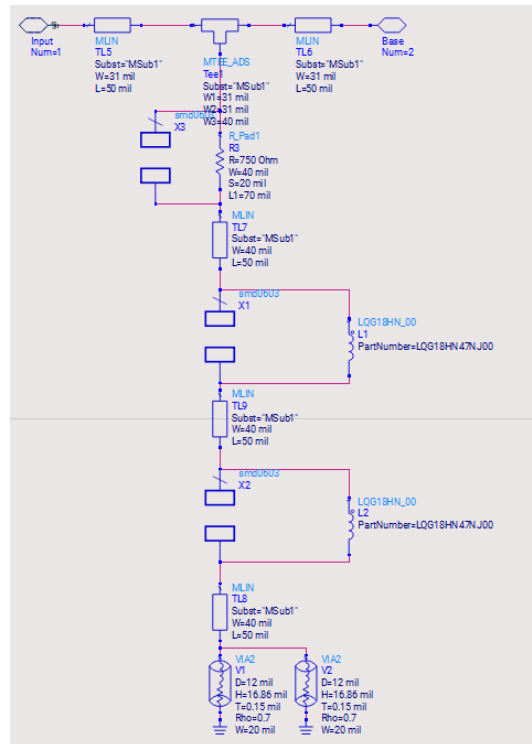
Since we noticed that our amplifier was already stable at our design frequency, we wanted to add the stabilizing components without lowering the gain at our desired frequency. To remediate the loss added to the system, we added self-resonant components where they would add effective shorts/opens where we wanted them at our design frequency. More specifically, for all shunt resistors added, we added a self-resonant inductor in series with the shunt resistor, which would present its maximum impedance at 1.3 GHz, reducing the effect of the resistor on our desired gain. Similarly, for all series resistances, we added a parallel self-resonant capacitor, which presents an effective short at our desired frequency.

The self-resonant components were selected based on the figure provided in the lab manual. For example, the capacitance graph is shown below, with our design frequency traced, showing that $\sim 40\text{pF}$ will be self-resonant at our frequency using these components.

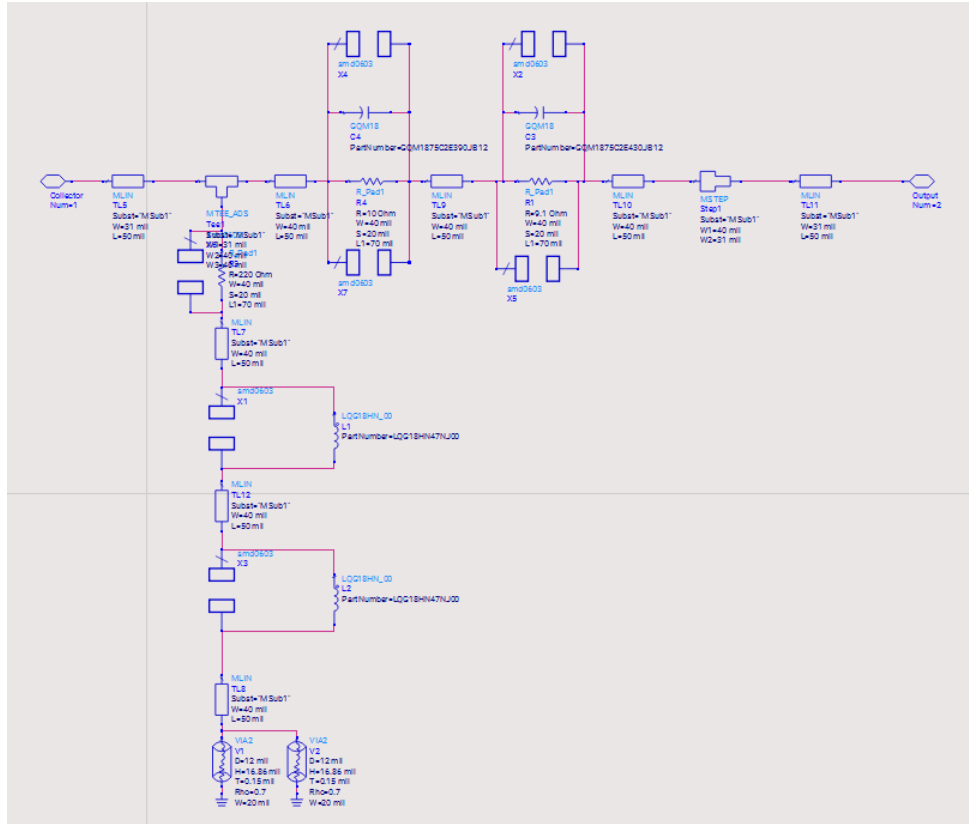


Self-Resonant Capacitor Chart for Murata Components

After iterating through idealized versions of these networks, adjusting components accordingly, we created the more physically realizable models for each network, which are shown in screenshots below, with their corresponding component values listed below them.



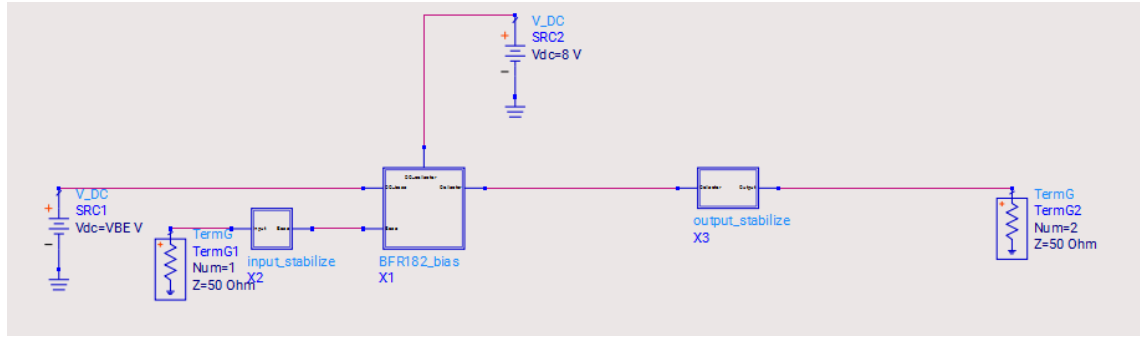
Input stabilization: Inductor (top & bottom) = 47 nH



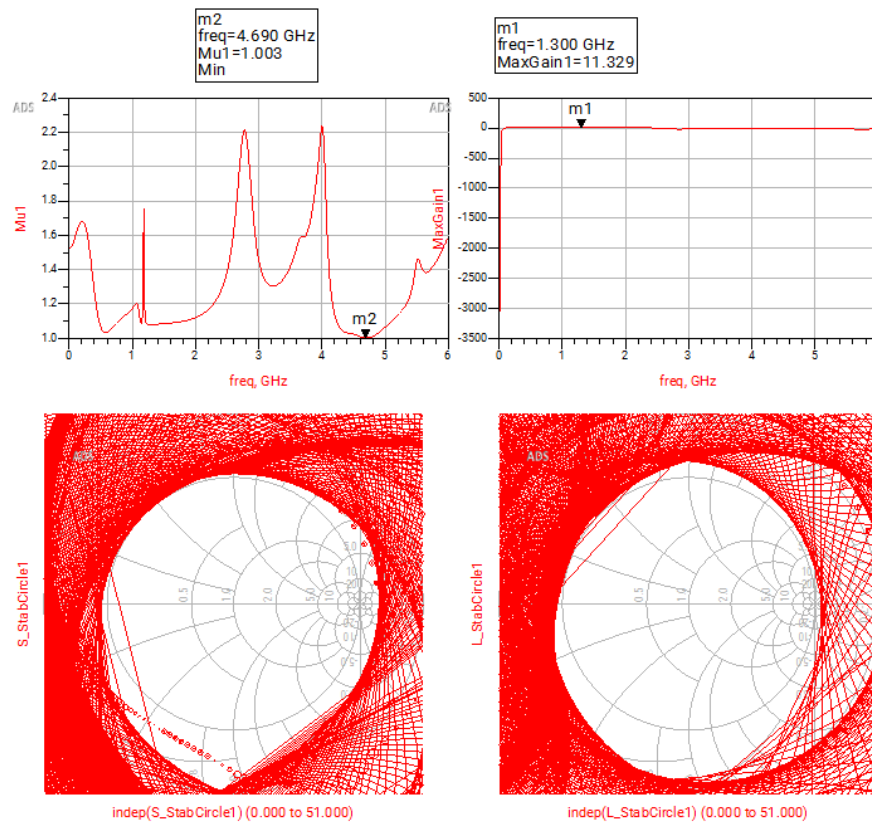
Output stabilization: Inductor(top & bottom) = 47 nH; Capacitor (left) = 39pF; Capacitor (right) = 43pF

In the output stabilization network, it can be seen above that we used two different capacitor values in parallel with two different resistors at the output. This idea stemmed from the fact that the 43pF capacitor in parallel with one resistor was leaving some surrounding frequencies conditionally unstable, as their gains were not being reduced much by this parallel combination. To remediate this, we split the resistor, now using two smaller resistors with different-valued capacitors in parallel with them. This created somewhat of a “bandpass” response that more selectively passed our desired frequency than the previous approach, resulting in stability surrounding it. These capacitor values both hover around the self-resonant values we found on the chart.

After adding the stabilization networks into the testbench, we tested again, measuring the stability and maximum gain over the frequency range. The testbench and results can be seen below. The mu stability factor remains above 1 for the entire range, and the maximum gain was 11.329 dB, which is around 0.7 dB higher than our target.



Testbench with Input and Output Stabilization



New Stability Results (Unconditionally Stable over Frequency)

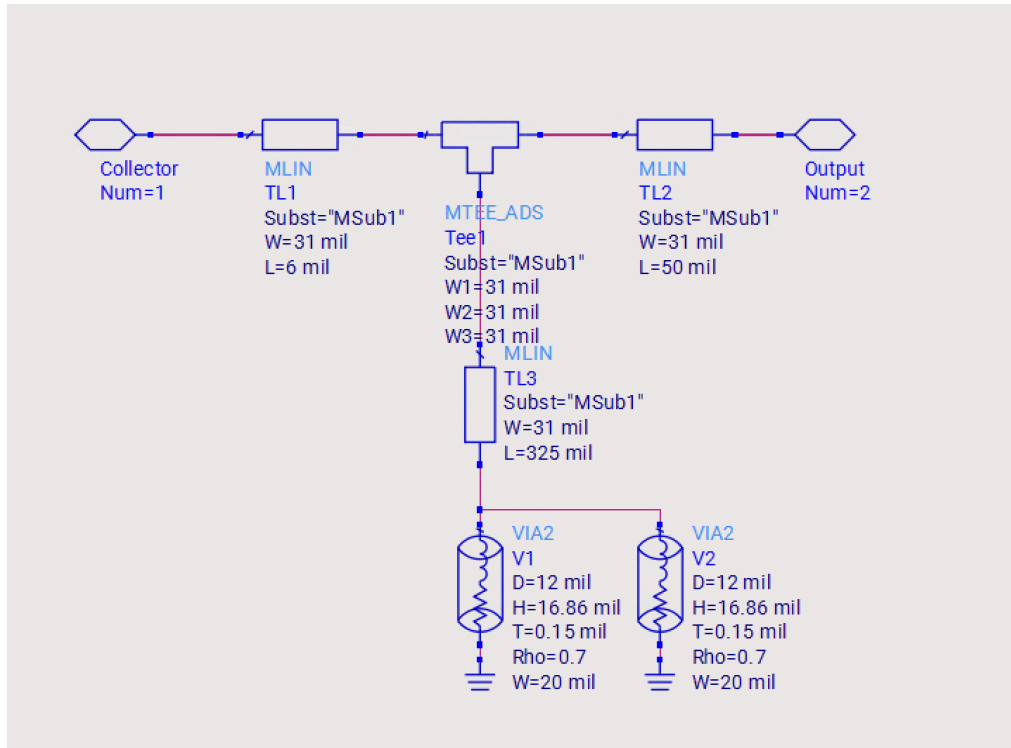
Impedance Matching

To begin the impedance matching phase of the design, we began by simulating and collecting the S-parameters at just our design frequency. These were the values we used to compute our simultaneous conjugate match values.

freq	S(1,1)	S(1,2)	S(2,1)	S(2,2)
1.300 GHz	0.003 - j0.166	0.091 - j2.462E-4	2.418 - j0.303	-0.527 - j0.509

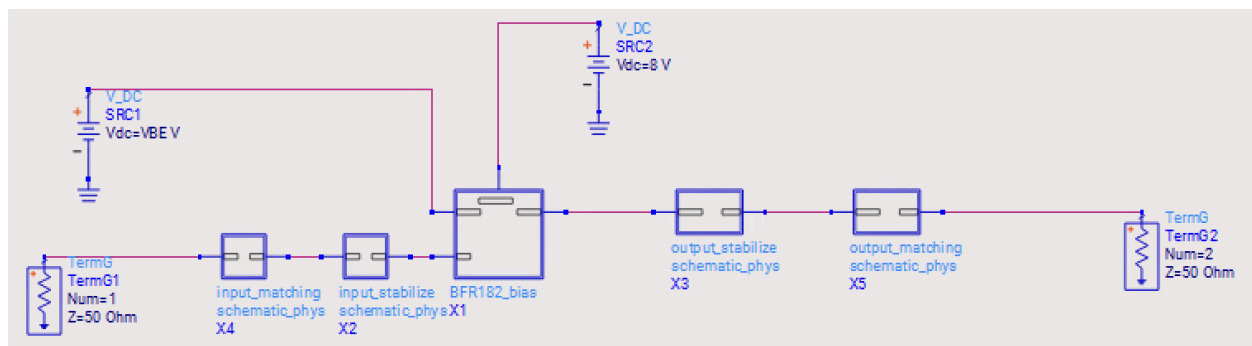
Initial S-Parameters After Stabilization

The screenshot below depicts the output matching network, which utilizes a distributed matching network, featuring a short-circuited stub. The values for these MLINs were determined using the Smith Chart tool to find the electrical lengths, then converted to physical length. From there, some fine-tuning was required to get the lengths just right.



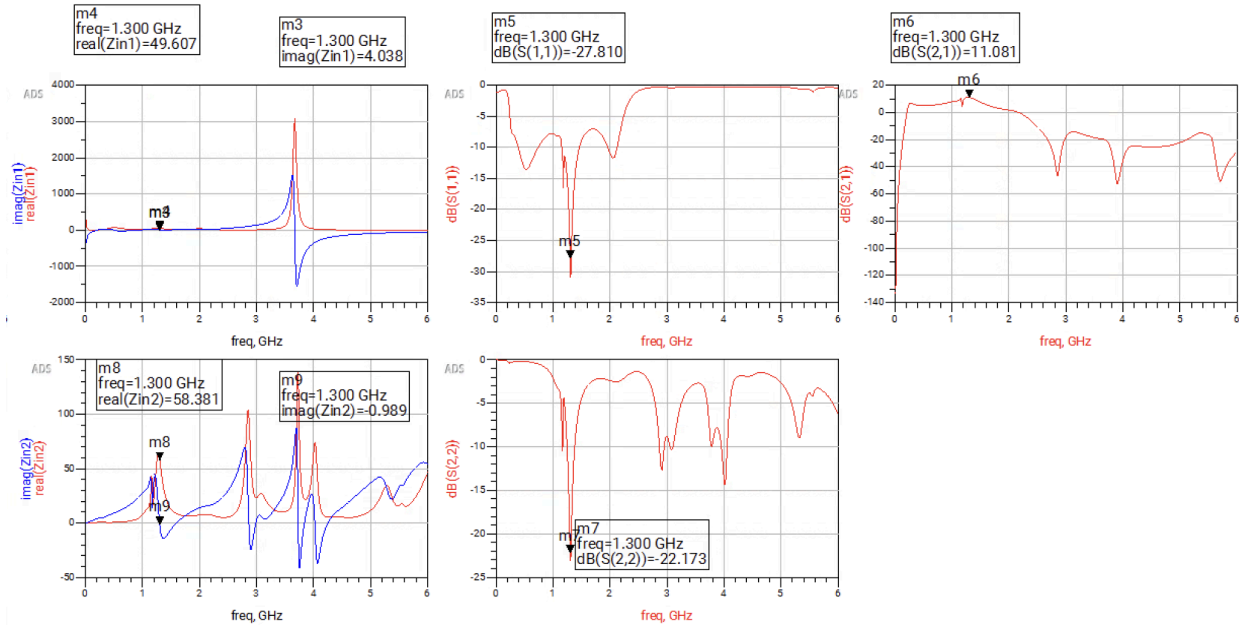
Output Matching Network Utilizing Distributed Components

After designing the matching networks and implementing them in physically realizable schematics, we attached them to our amplifier in the testbench shown below.



Testbench with Input and Output Matching Added

The results of this test were quite positive, with the amplifier now exhibiting 11.081 dB of forward gain (S_{21}) with S_{11} and S_{22} both being well under -20 dB. These can be seen in the screenshots below.



Results of Testing with Matching Networks

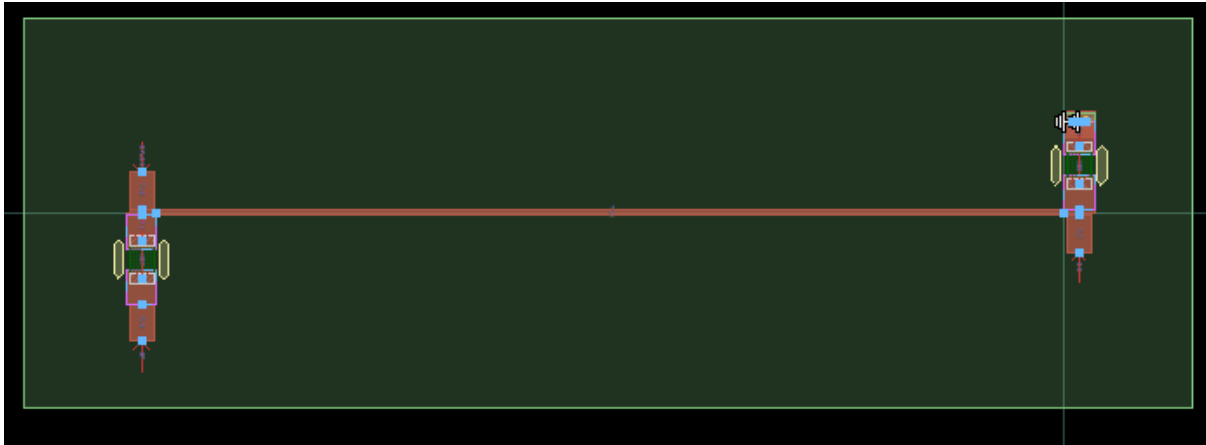
EM Co-simulation

After ensuring that the performance of our physically-implementable schematics (using realistic transmission line modeling) were up to the standard, it was time to use EM co-simulation to further prove the validity of our design. By using EM models of the bias tee and the matching networks, we were able to get a better understanding of how this design would perform on a real board.

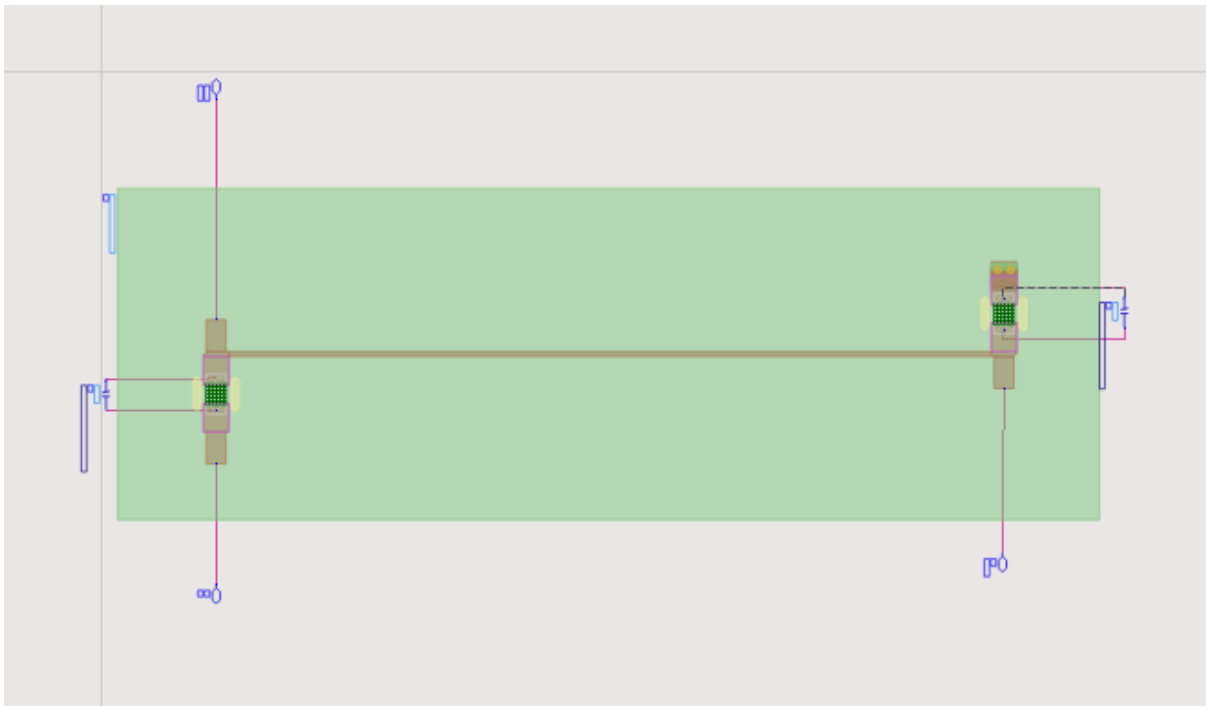
The first step in EM co-simulation is making the layout, which is made far simpler when the schematics are so accurate to how they should look in the final product (MTEEs are modeled, etc.). Most of these layouts can be largely generated automatically in ADS, which greatly accelerates the process.

The first layout to be completed was the bias tee. This was a straightforward process, with the only difficulty being properly aligning the edge of the ground plane with the ports of this block. This was made difficult by the fact that the capacitor on the DC side extends slightly further than the RF/DC output port does. We decided this wouldn't be too much of a problem, as we will be fixing this layout later when we will likely bend the quarter-wave transformer to conserve the

area. This layout can be seen below, with the co-sim schematic shown beneath it, showing how we added the Murata parts to the model.

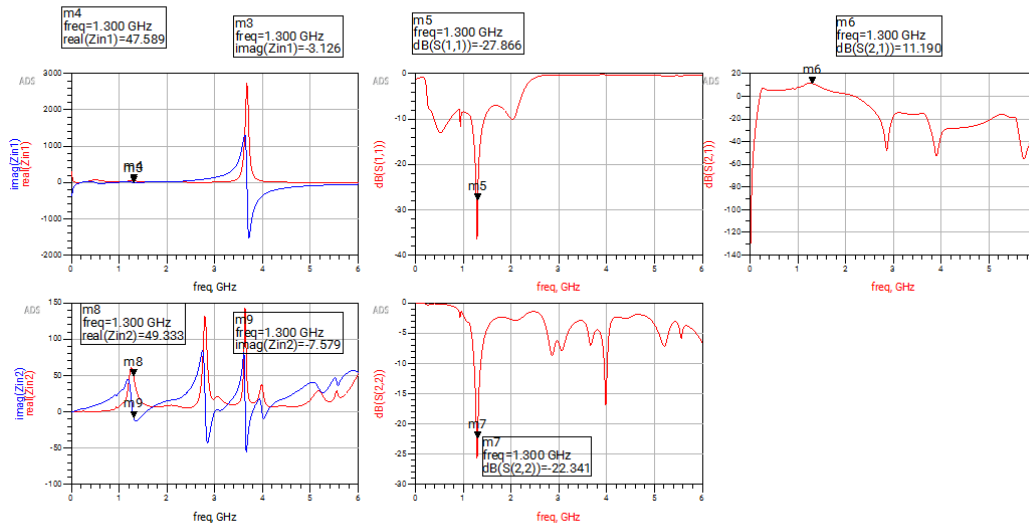


Layout of the Bias Tee



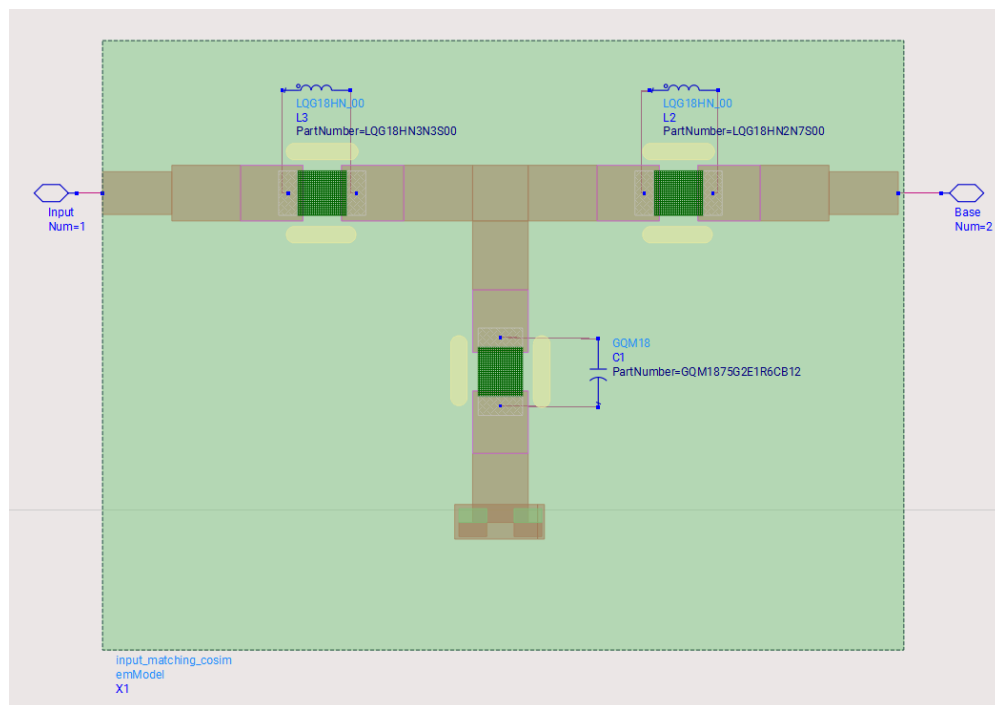
Co-simulation Model of the Bias Tee (with look-alike symbol)

After adding just the bias tee EM model to the simulation, we did not notice a major difference in the performance of the amplifier. Gain and impedance matching slightly improved with the addition of the EM model in fact. The results of this simulation can be seen in the following screenshot.

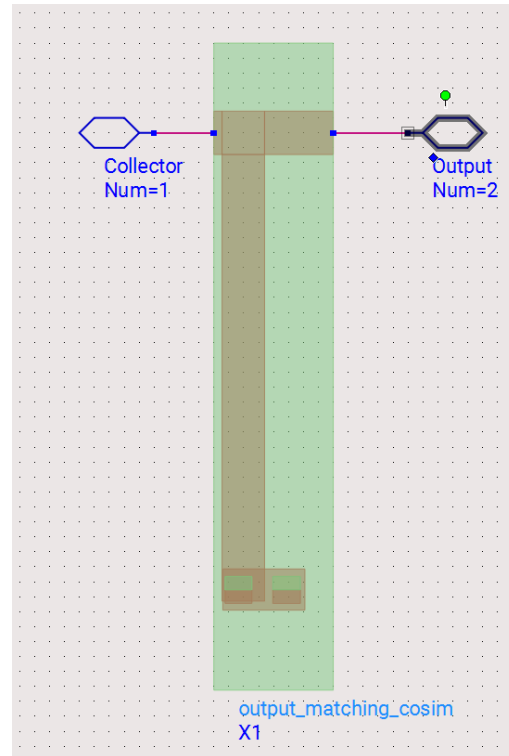


Testbench Results with Bias Tee EM Model Used

After the bias tee, we created the co-simulation models for both of the matching networks, by creating their layouts, then adding the Murata components to those schematics. Both of the matching networks with their layout-based symbols can be seen in the following screenshots.

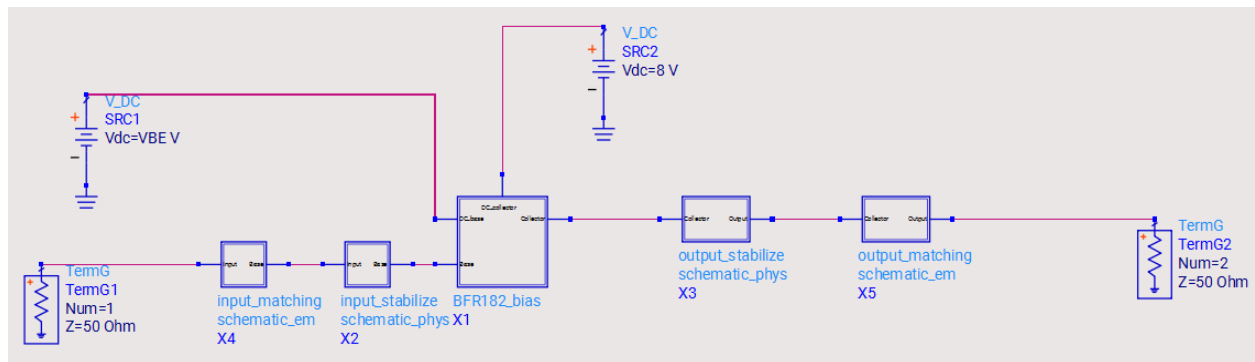


Input Matching Network Co-sim Model

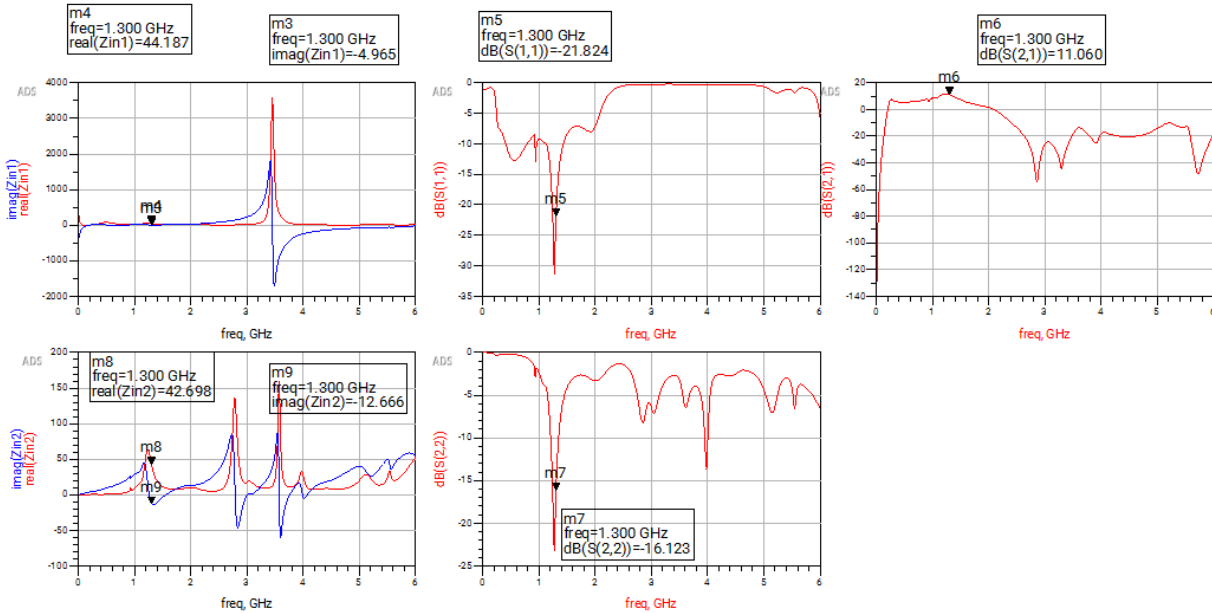


Output Matching Co-sim Model

With both of these EM models added, it was possible to simulate the entire amplifier with EM models for the bias tees and both of the matching networks. This would give us the most accurate insights on the performance of our amplifier that we could get without completing the entire layout or manufacturing the board. The testbench and the results can be found below.

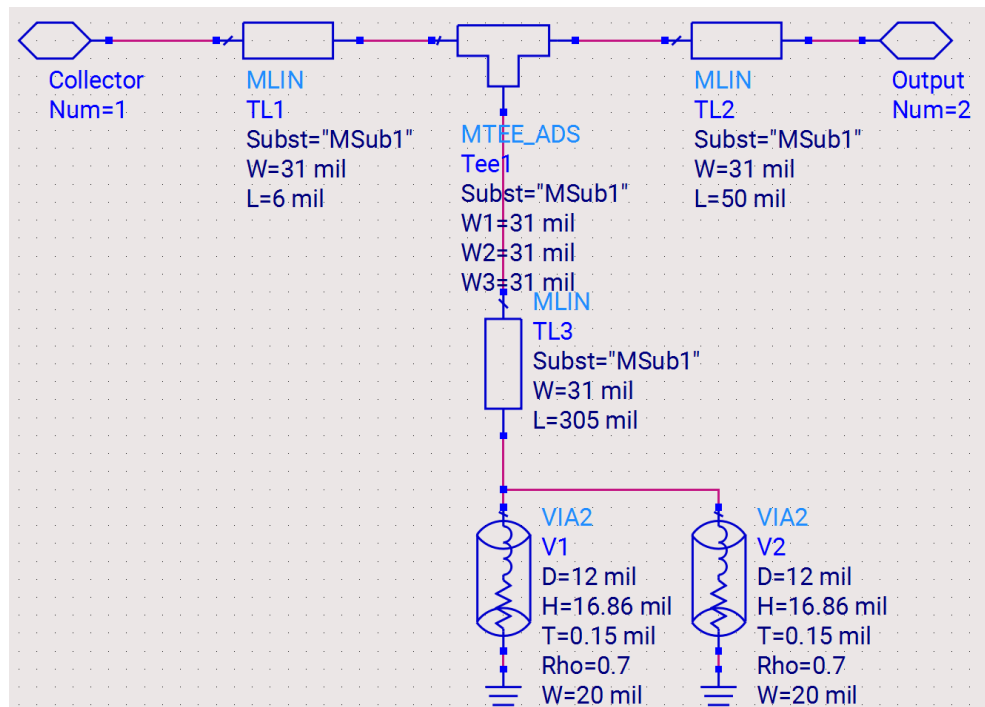


Testbench Utilizing EM Models of Bias Tee and Matching Networks



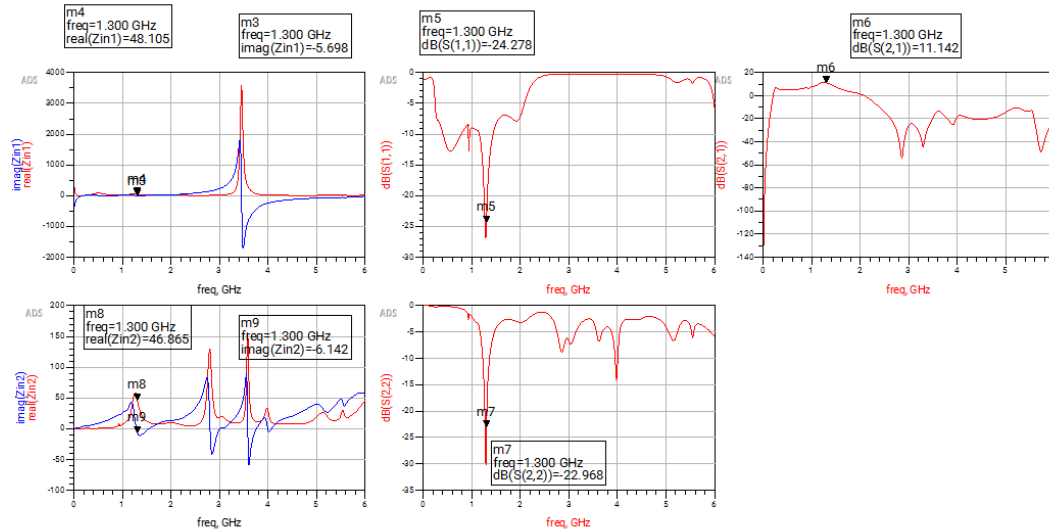
Initial Co-sim Results (slight dip in performance)

We noticed a dip in performance with the introduction of the matching network EM models, particularly in the S_{22} of the amplifier. It can be seen that the lowest point in the S_{22} plot had moved slightly to the lower frequency side. To fix this, we decided to decrease the length of the stub in the output matching network. The schematic and EM model were updated as shown below.



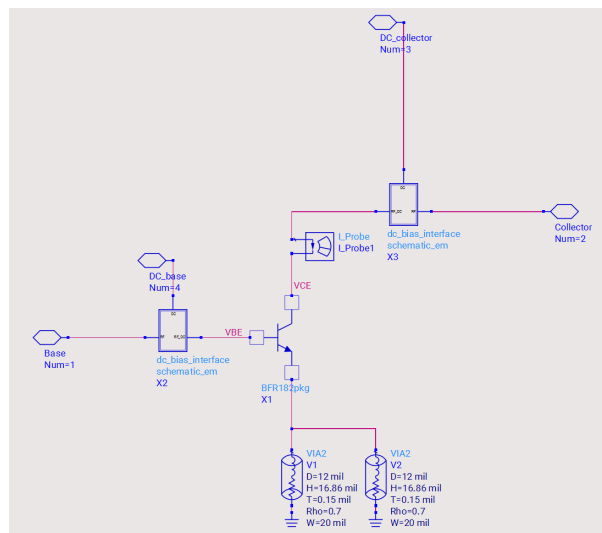
Improved Output Matching Network

The following co-simulation results are shown below. It can be seen that decreasing the length of the stub in the output matching network more effectively matched the load, meaning that our S_{22} (and S_{11}) improved, along with our gain (S_{21}) returning to its previous level.

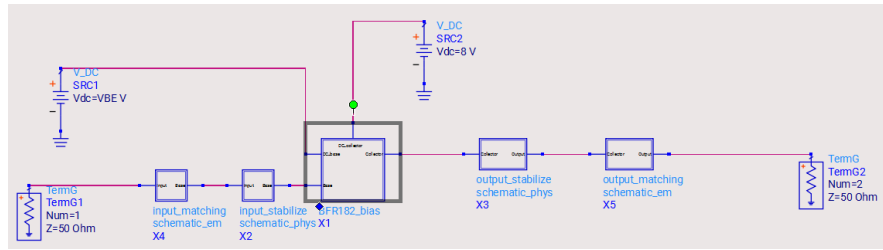


Improved Co-simulation Results

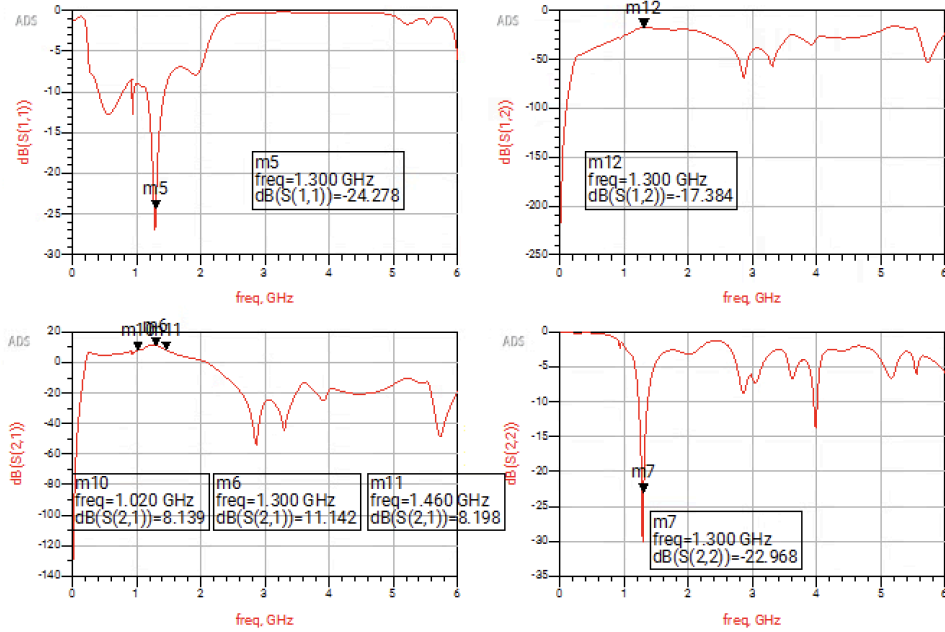
The screenshots below depict the current state of our hierarchical schematics, showing that the bias tees inside the “BFR182_bias” block have been replaced with their EM model/co-sim schematic, and the matching networks at the top-level have also been replaced with their EM model/co-sim schematics. This showcases the co-simulated amplifier and its results, which are in the following screenshots.



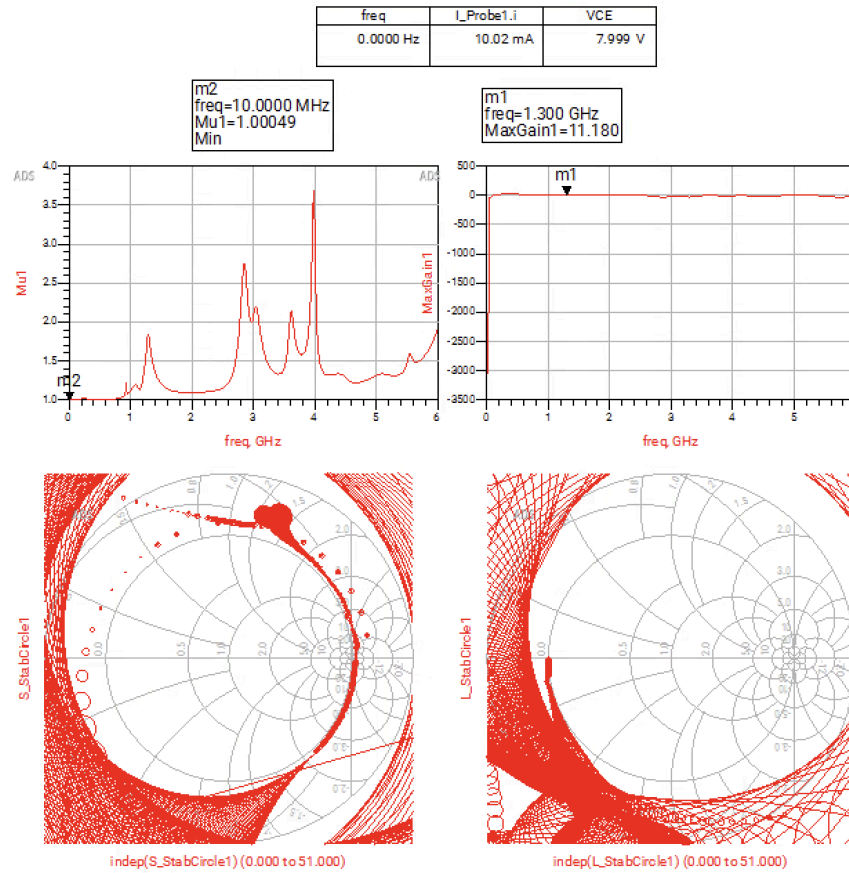
Co-simulation Model Shown in “BFR182_bias” sub-circuit



All Co-sim Models on the Top-Level Testbench



Amplifier S-Parameters with Co-sim Bias Tee and Matching Networks



Stability and DC Bias Point with Co-sim Bias Tee and Matching Networks

Our amplifiers performance is listed in the following table:

Parameter	Specification	Value
S ₁₁	-15 dB	-24.278 dB
S ₁₂	N/A	-17.384 dB
S ₂₁ (gain)	10.6 dB	11.142 dB
S ₂₂	-15 dB	-22.968 dB
Fractional Bandwidth	5%	33.85%
Stability (μ ; 10MHz - 6 GHz)	1	1.00049
V _{CE}	8 V	7.999V
I _C	10 mA	10.02 mA